ModelIt!

Student Activity Pack

Chain Reaction: When Energy Moves Object to Object

Why Each Collision Loses a Little Energy Along the Way

Grade 04L2 | NGSS 4-PS3-3, 4-PS3-4 | Energy and Matter

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved.

Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

9 Post-Assessment 11

1 Pre-Assessment

ModelIt!

What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 When you knock over a line of dominoes, what do you think makes each domino fall?

2 If you roll a marble into a line of marbles, why does the last marble shoot out the other end?

3 Draw what you think happens to the ENERGY when a moving ball hits a stationary ball.

4 Why do you think you can HEAR collisions? Where does that sound come from?

2 Scoring Rubric

ModelIt!

Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

ModelIt!

What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Initial Push Force | Object Mass | Kinetic Energy Transfer | Energy Lost to Sound | Final Object Speed

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Strong Push, Light Objects

Initial Push Force at 80%, Object Mass at 20%

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Weak Push, Heavy Objects

Initial Push Force at 20%, Object Mass at 80%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Strong Push, Heavy Objects

Initial Push Force at 80%, Object Mass at 80%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Number of Objects in Chain — How many objects the energy must pass through — more objects means more collisions and more opportunities for energy loss

Object Material — What the objects are made of — rubber balls bounce and transfer energy differently than steel balls or wooden blocks

Collision Angle — Whether the objects hit each other straight on or at an angle — straight-on collisions transfer energy more efficiently

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

ModelIt!

The Big Question

If you set up a line of dominoes and push the first one, why does the last domino always fall slower than the first one was pushed? Use evidence from your simulation to support your answers.

1. **Why does each collision in a chain reaction have slightly less energy than the one before it?**

2. **If you can hear a collision, what does that tell you about where some energy went?**

3. **How is a Newton's cradle different from a line of dominoes in terms of energy transfer?**

4. **If you wanted to send energy through the longest possible chain, would you use heavy or light objects? Why?**

5. **How did your model help you understand why chain reactions eventually run out of energy?**

8 STEM Challenge: Build the Longest Energy Chain

ModelIt!

YOUR ENGINEERING MISSION

Design a chain reaction where energy transfers through as many objects as possible while still having enough energy to complete the final action.

THE SCENARIO

A science museum wants a Rube Goldberg machine exhibit that shows visitors how energy transfers from object to object. Your engineering team must build the longest chain reaction possible where a push at the start causes a final action at the end — like ringing a bell, knocking over a cup, or launching a small ball. The chain must have at least five collisions, and you must explain where energy is lost at each step.

Engineering Design Thinking

1. WHAT OBJECTS WILL YOU USE IN YOUR CHAIN (MARBLES, DOMINOES, TOY CARS, BALLS, BLOCKS)?

2. HOW WILL YOU ARRANGE THEM SO EACH COLLISION TRIGGERS THE NEXT?

3. WHAT WILL YOUR FINAL ACTION BE — WHAT DOES THE LAST OBJECT DO TO PROVE ENERGY MADE IT THROUGH?

4. HOW WILL YOU MINIMIZE ENERGY LOSS AT EACH COLLISION TO MAKE THE CHAIN AS LONG AS POSSIBLE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Mechanical Engineers and Physicists study how energy transfers between objects to design everything from car safety systems (crumple zones absorb collision energy) to sports equipment (bats and clubs transfer energy efficiently to balls). They earn $70,000–$120,000/year.

9 Post-Assessment

ModelIt!

Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 When a moving marble hits a row of stationary marbles, the last marble in the row rolls away. What happened to the energy?

- **A.** The moving marble created new energy that appeared in the last marble
- **B.** The kinetic energy from the first marble transferred through the chain from marble to marble until it reached the last one
- **C.** The last marble moved on its own because it saw the first marble coming
- **D.** Gravity pulled the last marble forward when the first marble stopped

2 In the model, Initial Push Force and Object Mass are external components. Why are these external?

- **A.** Because they are the parts we cannot see in the chain reaction
- **B.** Because they are the starting conditions we control, and the internal components respond to them
- **C.** Because they happen after the chain reaction is finished
- **D.** Because they are the parts of the model that change the most

3 Every time one domino hits the next in a chain, you hear a clicking sound. What does that clicking sound tell you about energy?

- **A.** The sound means new energy is being created at each collision
- **B.** The sound is proof that some kinetic energy is transforming into sound energy at each collision, reducing the energy available for motion
- **C.** The sound has nothing to do with energy
- **D.** The sound means the dominoes are gaining energy as they fall

4 The model shows that Object Mass has a NEGATIVE relationship with Final Object Speed. A student tests the chain with light marbles and then with heavy steel balls. What will they observe?

- **A.** Heavy objects will make the last object move faster because they hit harder
- **B.** Both chains will have the same Final Object Speed because the push was the same
- **C.** The light marble chain will have a faster Final Object Speed because lighter objects transfer energy more efficiently and require less energy to get moving
- **D.** Neither chain will work because objects cannot transfer energy

5 **The model shows that Energy Lost to Sound has a NEGATIVE relationship with Final Object Speed. Why does a long chain of 20 dominoes produce a slower final domino than a short chain of 5 dominoes with the same starting push?**

- **A.** The dominoes get tired after falling for too long
- **B.** A longer chain has more collisions, and each collision loses energy to sound and heat, so less energy reaches the final domino
- **C.** The first domino in the long chain pushes harder than the first domino in the short chain
- **D.** Longer chains create friction between the dominoes and the table